IMPLEMENTATION OF FASTSPEECH 2-BASED TEXT-TO-SPEECH SYNTHESIS

for English and Telugu

Technical Report

Submitted by

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GitHub Repository:
`https://github.com/vishakhmv/fastspeech2-english-telugu-tts`

Checkpoint Folder:
`https://drive.google.com/drive/folders/1KYR0W8YtH0LEsdu4d1INNzAjR7Ino1uo`

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Abbreviations

AMP	–	Automatic Mixed Precision
ASR	–	Automatic Speech Recognition
F0	–	Fundamental Frequency (pitch)
FFT	–	Feed-Forward Transformer
G2P	–	Grapheme-to-Phoneme conversion
GPU	–	Graphics Processing Unit
HiFi-GAN	–	High-Fidelity Generative Adversarial Network (vocoder)
MFA	–	Montreal Forced Aligner
MSE	–	Mean Squared Error
PER	–	Phone Error Rate
PESQ	–	Perceptual Evaluation of Speech Quality
POS	–	Part of Speech
SI-SDR	–	Scale-Invariant Signal-to-Distortion Ratio
SNR	–	Signal-to-Noise Ratio
SQUIM	–	Speech Quality and Intelligibility Measures (TorchAudio)
STFT	–	Short-Time Fourier Transform
STOI	–	Short-Time Objective Intelligibility
TTS	–	Text-to-Speech
WER	–	Word Error Rate

Abstract

This report describes how FastSpeech 2 was implemented and trained as two separate text-to-speech models, one for English and one for Telugu. The open-source PyTorch implementation by ming024 was the starting point. Its code was reorganised into one shared package, with separate scripts and configuration files for each language.

Most of the new work went into Telugu, which the reference code does not support at all. A rule-based grapheme-to-phoneme front-end with number normalisation was written for it, and a forced-alignment model was trained on the Telugu recordings. The English corpus, Svarah, is accented and noisy, so its clips were ranked by estimated quality before the training data was chosen. Pitch is normalised per speaker, and pitch and energy are embedded as continuous values instead of quantised bins.

Each model was trained on 5 hours of speech and validated on 1 hour, and the two sets share no sentence. The English model covers 114 Indian-accented speakers from Svarah; the Telugu model covers the male speaker of IndicTTS Telugu. Both models converged within about 10,000–16,000 steps. After that the decoder stayed flat, while the duration, pitch and energy predictors overfitted slightly.

Intelligibility was measured with Whisper speech recognition on unseen validation sentences. The Telugu model reaches a phone error rate of 6.9%, against 6.0% for the real recordings, and the English model a word error rate of 12.6%, against 9.9%. On the ten test sentences of the assignment, the recogniser's only errors are a hyphenated spelling, the spelling of the name "andaroo" and one lengthened vowel in a Telugu loanword.

The variance-adaptor plots show pitch falling over almost every sentence, longer phones at the ends of phrases, pauses only at commas, and correctly kept Telugu vowel length and doubled consonants. They also show the main difference between the two languages: the English pitch predictor explains only 11% of the validation pitch variation, against 47% for Telugu, so English intonation comes out much flatter.

Keywords: text-to-speech, FastSpeech 2, variance adaptor, Telugu, Indian-accented English, HiFi-GAN.

Chapter 1

Introduction

Text-to-speech (TTS) systems turn written text into speech. Most neural TTS systems do this in two steps. An acoustic model first predicts a mel-spectrogram from the phoneme sequence, and a vocoder then converts the mel-spectrogram into a waveform.
FastSpeech 2 [1] is a non-autoregressive acoustic model, which means that it generates all mel frames at once rather than one after another. It does not learn the alignment between text and speech through attention. Instead, a *variance adaptor* predicts the duration, pitch (F0) and energy of every phoneme, and these predictors are trained on values measured from real recordings. This makes synthesis fast and stable. It also means that the prosody of each generated sentence can be inspected and changed directly, which is what the analysis in Chapter 6 relies on.

1.1 Objective

The assignment asks for FastSpeech 2 to be implemented and evaluated separately for English and Telugu, starting from an existing GitHub implementation. The main objectives were:

- to adapt the reference code to both languages, with particular attention to Telugu support;
- to train one model per language on the specified datasets, with about 5 hours of training audio and 1 hour of validation audio each;
- to generate speech for the five test sentences of each language, keeping the text exactly as given;
- to analyse the duration, pitch and energy that the variance adaptor predicts for those sentences.

1.2 Scope and Deliverables

The English model is trained on Svarah [4], a corpus of Indian-accented English, and the Telugu model on IndicTTS Telugu [5]. The repository contains the complete preprocessing, training and inference code, a configuration file and a checkpoint for each language, the training and validation loss plots, the generated audio for the test sentences, and duration, pitch and energy plots for every test sentence.
This report covers the implementation, the data, the training behaviour, the generated speech, the analysis of the variance adaptor, a comparison of the two languages, and the steps needed to reproduce the results.

1.3 Summary of Results

Both models trained without instability. The validation loss of the decoder stopped improving after about 10–16k steps, while the variance predictors started to overfit slightly (Chapter 4). On unseen validation sentences the Telugu model is close to real recordings, with a phone error rate of 6.9% against 6.0%. The English model is further away, with a word error rate of 12.6% against 9.9% (Chapter 5).
The test sentences contain heteronyms, numbers, dates, times and deliberately broken words, and all of them are synthesised from the text exactly as given. The released English voice makes no real word error according to the recogniser, and the Telugu voice gets 1 of 284 phones wrong. By ear, however, some fast parts of the English sentences are harder to follow than these scores suggest (Section 5.4).
The variance adaptor reproduces the expected prosodic patterns: pitch that falls over the sentence, longer phones at the ends of phrases, pauses at commas and Telugu doubled consonants. English intonation is much flatter than Telugu intonation, and Chapters 6 and 7 discuss why.

Chapter 2

Implementation Approach and Architecture

2.1 Approach

The PyTorch implementation by ming024 [2] was used as the starting point. Its main building blocks were kept: the feed-forward Transformer (FFT) blocks, the variance predictors, the length regulator, the PostNet, the loss functions and the scheduled optimiser. The pretrained HiFi-GAN universal vocoder [3] was kept as well.

The code itself was reorganised. Everything that the two languages share, namely the model, the training loop, feature extraction, inference and plotting, now exists only once, in the package `fastspeech2/`. The folders `English/` and `Telugu/` hold the language-specific `preprocess.py`, `train.py`, `inference.py` and `config.yaml`, together with the checkpoints. As a result, the two models differ only in their configuration, text front-end, alignment setup and data.

2.2 Model Architecture

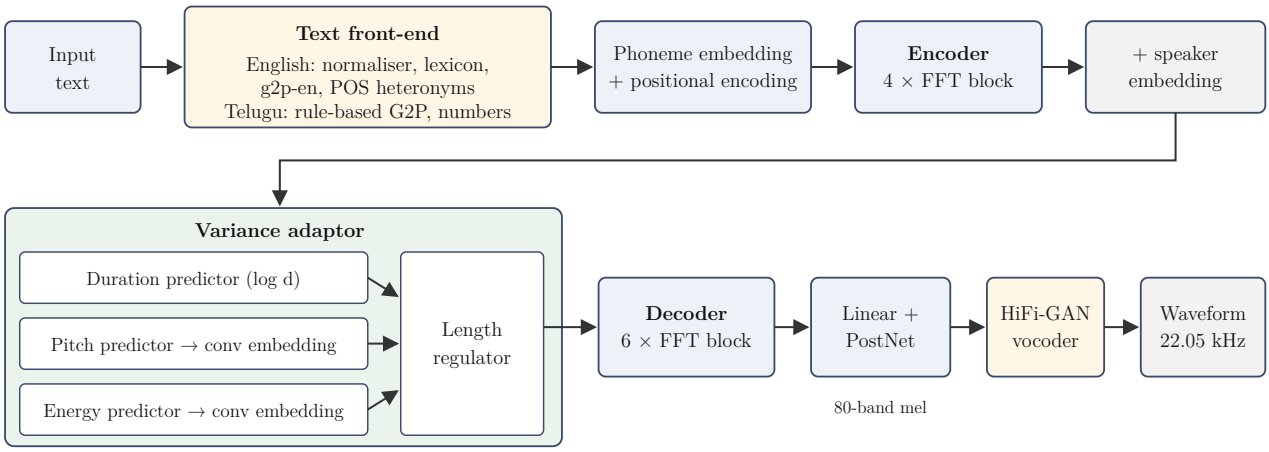


Figure 2.1: FastSpeech 2 synthesis path as implemented

Figure 2.1 shows the synthesis path. A language-specific front-end converts the input text to phonemes, which are embedded and passed through four FFT blocks. A learned speaker embedding is added to the encoder output. The variance adaptor then predicts a log-duration, a pitch value and an energy value for each phoneme. Pitch and energy are embedded and added back to the hidden sequence, and the length regulator repeats each phoneme state for its predicted number of frames. Six decoder FFT blocks and a linear layer produce an 80-band mel-spectrogram. The PostNet refines it, and HiFi-GAN turns it into a 22,050 Hz waveform. The orange boxes in the figure are the parts that are specific to a language or come from outside the model.

Table 2.1 lists the components and their settings, which are the same for both languages. Each model has about 34.5 M trainable parameters.

Component	**Configuration**
Phoneme encoder	4 FFT blocks, hidden size 256, 2 attention heads, convolution filter size 1024 with kernels [9, 1], dropout 0.2
Variance predictors	Separate duration, pitch and energy predictors, each with two 1-D convolution layers (256 filters, kernel 3), ReLU, LayerNorm and dropout 0.5, followed by a linear output
Pitch and energy embedding	1-D convolution (kernel 3) over the normalised per-phoneme values, added to the hidden sequence
Length regulator	Repeats each phoneme state for its duration in frames (hop of 256 samples, 11.6 ms)
Mel decoder	6 FFT blocks, hidden size 256, 2 attention heads
PostNet	5 convolution layers, 512 channels, kernel 5; residual refinement of the 80-band mel-spectrogram
Speakers	Learned speaker embedding added to the encoder output: 114 English speakers, 4 Telugu session IDs
Loss	L1 on the mel-spectrogram before and after the PostNet, plus MSE on phoneme-level pitch, energy and log(duration + 1)
Vocoder	Pretrained HiFi-GAN universal generator [3] at 22,050 Hz, not fine-tuned

Table 2.1: Architecture components and configuration

2.3 Modifications for English and Telugu

Table 2.2 compares the reference implementation with the version used in this work. The two largest changes, Telugu support and the English text front-end, are described in more detail after the table.

Area	Reference implementation	This work
Telugu text front-end	Not supported	New rule-based grapheme-to-phoneme converter, Telugu phone set and Telugu number normaliser (Section 2.3.1)
English text front-end	Lexicon lookup with g2p-en fallback and basic text cleaners	Normalisation of times, dates, years, ordinals, decimals, acronyms and symbols; ARPAbet lexicon from the MFA `english_us_arpa` dictionary; g2p-en [10] for unknown words; heteronyms resolved by part of speech
Forced alignment	Pre-computed MFA TextGrids for its own datasets	Montreal Forced Aligner [6]: the pretrained English model for Svarah and a new acoustic model trained on the Telugu data
Data selection	Uses complete corpora	Quality and alignment ranking for Svarah; 5 h / 1 h splits without shared sentences for both languages (Chapter 3)
Pitch extraction	WORLD pitch with global normalisation	Praat F0 [9] with a search range per speaker; log-F0; normalisation per speaker using training statistics only
Pitch and energy embedding	256 quantised bins	Continuous convolutional embedding of the normalised values
Encoder gradients	Full gradient from the variance predictors	Gradient from the predictors into the encoder scaled by 0.1, so the predictors reshape the encoder less
Speakers	Speaker embedding only for the multi-speaker corpora (AISHELL-3, LibriTTS)	114 English speakers; the Telugu voice split into 4 recording-session IDs
Training	900k steps in full precision	60k (English) and 66k (Telugu) steps; mixed precision; automatic resume and time limits for Colab and Kaggle; `best.pt` chosen by the validation mel loss with predicted pitch and energy
Inference and plots	Synthesis with global control factors	Voice selection; separate duration scales for speech and pauses; duration, pitch and energy plots per sentence with measured contours; `inference_details.json`

Table 2.2: Modifications relative to the reference implementation

2.3.1 Telugu Support

Telugu is written in an abugida, a script in which every consonant letter carries an inherent vowel, and its spelling follows the pronunciation closely. A rule-based converter is therefore

enough to go from letters to phonemes. The front-end in `fastspeech2/text/telugu.py` applies the following rules:

- every consonant carries the inherent vowel /a/, unless a dependent vowel sign replaces it or a virama (◌్) removes it;
- a doubled consonant becomes two identical phones, so పట్టు becomes `p a T T u`;
- an anusvara becomes the nasal that matches a following velar, palatal, retroflex or dental consonant (ṅ, ñ, ṇ or n), and m everywhere else;
- a visarga becomes /h/.

The resulting phone set has 43 phones plus a pause symbol. Numbers are written out as Telugu words, including years, ordinals with the suffix వ and clock times (Table 5.4). The pronunciation lexicon for the aligner is generated with the same rules, so training and inference always use the same pronunciations.

2.3.2 English Front-End

English text is normalised before any lookup. For example, “10:45 AM” becomes *ten forty five A M* and “2026” becomes *twenty twenty six*. Each word is then looked up in an ARPAbet lexicon with stress marks; all 69 stressed ARPAbet phones occur in the data, plus a pause symbol. Words that are missing from the lexicon are looked up in the CMU Pronouncing Dictionary that comes with g2p-en, and only words that are not there either, such as “andaroo”, are predicted by the g2p-en model. A small rule reads a final “-oo” spelling as `UW1`.

Heteronyms are resolved with part-of-speech tags. The noun “conduct” in E2a becomes `K AA1 N D AH0 K T`, while the verb in E2b becomes `K AH0 N D AH1 K T`. The front-end also contains an optional step that joins broken words such as “rhyth m” back together. It is switched off for the deliverables, because the assignment asks for the test sentences to be kept exactly as provided.

Chapter 3

Datasets and Preprocessing

3.1 Datasets

Table 3.1 summarises the two corpora, which are very different from each other. Svarah was recorded as a test set for English speech recognition. It contains speakers from many parts of India, recorded under varying conditions. IndicTTS Telugu, on the other hand, is a studio corpus made for speech synthesis.

Language	**Dataset**	**Content**	**Part used**
English	Svarah [4] (CC BY 4.0)	6,656 clips of Indian-accented English from many speakers, with speaker metadata; designed as a test set for speech recognition	The best-quality 6 h: 4,037 clips from 114 speakers
Telugu	IndicTTS Telugu [5] (Hugging Face parquet release)	8,576 studio clips of read Telugu from one male and one female speaker	The 4,533 clips of the male speaker

Table 3.1: Datasets

The Telugu speaker labelled as male reads with a fairly high voice, with an average F0 of 167–197 Hz depending on the recording session. His second and third formants are 15–18% lower than those of the female speaker, which is consistent with the male label. This is why the Telugu plots show F0 values around 200 Hz.

3.2 Data Selection and Subsampling

English. The recording quality in Svarah varies a lot, so the clips were ranked by quality before the 6-hour budget (5 h for training and 1 h for validation) was filled. The selection had four steps:

1. **Hard filters.** A clip had to last 1–18 s, have a speaking rate of 0.6–4.6 words per second and a transcript that could be normalised, contain at most 0.2% clipped samples,

and have at least 20% voiced frames.

2. **Quality score.** Each clip was scored with the reference-free TorchAudio-SQUIM estimators [8] and two simple signal measures, weighted 0.35 for SI-SDR, 0.25 for PESQ, 0.15 for STOI, 0.15 for an SNR estimate and 0.10 for a high-frequency ratio. The worst 12% of the clips were dropped.

3. **Alignment filter.** After forced alignment, clips below the 3rd percentile of alignment log-likelihood were removed.

4. **Final ranking.** The remaining clips were ranked by their quality score plus 0.5 times the standardised alignment likelihood, and the best 6 hours were kept.

Telugu. The male recordings are clean, so simpler filters were enough. A clip had to last 1.5–16 s, contain 7.5–16.5 characters per second and have at most 0.1% clipped samples, and clips below the 5th percentile of alignment likelihood were removed.
The male voice was recorded over several sessions, and the sessions sound slightly different. To keep these differences from blurring the voice, the utterance indices were divided into six ranges, which were grouped into four session IDs, `male_a` to `male_d`. The largest of them, `male_a`, holds 150 of the 300 training minutes.
Table 3.2 shows how many clips were left after each step. For Telugu, the length, rate and clipping filters are applied together in one step. The hours before alignment are clip durations, while the last row uses the aligned durations with silences trimmed.

Stage	**English**	**Telugu**
Source clips	6,656	4,533 (male)
After duration, rate and text filters	4,805 (7.46 h)	3,647 (8.21 h)
After clipping and voicing checks	4,795 (7.45 h)	–
After dropping the worst 12% by quality	4,219 (6.63 h)	–
After alignment filter and 6 h budget	4,037 (5.99 h)	3,090 (6.00 h)

Table 3.2: Data selection funnel

3.3 Train/Validation Split

Both splits are text-disjoint. Clips are grouped by their normalised sentence before splitting, so the same sentence can never appear in both sets.
For English, every speaker with at least 90 s of audio gives about one sixth of their audio to validation. In this way the validation set covers 76 of the 114 speakers, while most of each speaker's data still goes to training. For Telugu, the sentence groups were shuffled with seed 1234 and assigned until there was 1 h of validation audio and 5 h of training audio.
A check after the split confirmed that the two sets share no clip and no sentence in either language. Table 3.3 gives the final sizes.

Language	Split	Clips	Hours	Mean clip (s)	Speakers
English	Train	3,391	4.99	5.30	114
English	Validation	646	1.00	5.56	76
Telugu	Train	2,570	5.00	7.01	4 session IDs
Telugu	Validation	520	1.00	6.93	4 session IDs

Table 3.3: Final train and validation splits (aligned speech, silences trimmed)

3.4 Preprocessing and Feature Extraction

Figure 3.1 shows the preprocessing stages of both languages. English has an extra analysis stage that computes the quality scores, and Telugu has to train its own aligner model before it can align.

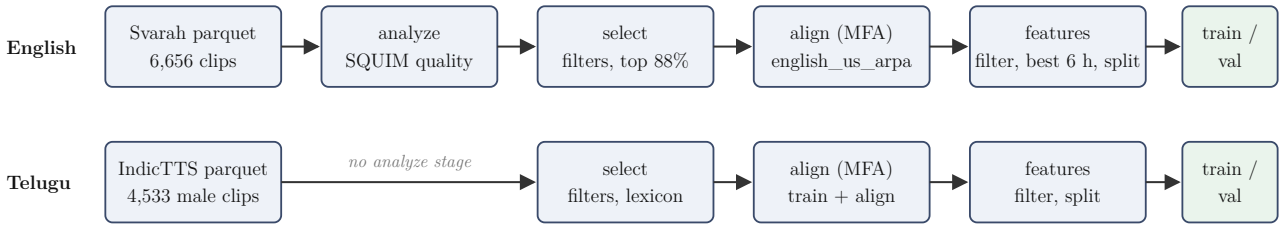


Figure 3.1: Preprocessing stages for English and Telugu

The individual steps are the following.

- **Audio.** All clips are resampled to 22,050 Hz and normalised to an active speech level of −23 dB with a peak limit of 0.95. A 16 kHz copy is kept for alignment.
- **Alignment and durations.** The transcripts are aligned with the Montreal Forced Aligner 3.4.1 [6]. English uses the pretrained `english_us_arpa` model (beam 100, retry beam 400). For Telugu, a new acoustic model is trained on the selected corpus. Phone durations are the aligned frame counts.
- **Mel-spectrograms.** 80 mel bands between 0 and 8 kHz, with a 1024-point Fourier transform, a hop of 256 samples and a window of 1024 samples.
- **Pitch.** F0 is extracted with Praat through Parselmouth [9]. Each speaker gets their own search range: a first pass uses 50–700 Hz, and a second pass uses 0.65 × Q25 to 2.2 × Q75 of that speaker's F0. Log-F0 is interpolated through unvoiced frames, averaged per phoneme and z-normalised per speaker with training-set statistics only.
- **Energy.** The frame energy of the STFT magnitude, averaged per phoneme and z-normalised per speaker.

Chapter 4

Training and Loss Analysis

4.1 Training Configuration

Both models are trained with the Adam optimiser and the learning-rate schedule of the reference implementation,

$$lr = 256^{-0.5} \cdot \min(step^{-0.5}, step \cdot 4000^{-1.5}) \cdot 0.3^{k},$$

where k is the number of decay steps already passed. The reference schedule runs for 900k steps, far more than free GPU sessions allow. In the first part of the Telugu run the validation loss stopped improving after about 16k steps, so the schedule was shortened to 60k steps for English and 66k for Telugu, with the decay steps placed in the last part of training (Table 4.1).

Training ran on free NVIDIA T4 GPUs. English was trained in one Kaggle session of about 3.6 hours. The Telugu run started on Google Colab, which stopped it at step 44,800 when the free GPU time ran out; it was resumed on Kaggle from the checkpoint saved at step 44,000.

Setting	**English**	**Telugu**
Optimiser	Adam ($\beta = 0.9, 0.98$; $\varepsilon = 10^{-9}$), 4,000 warm-up steps, gradient clipping at 1.0	
Learning-rate decay ($\times 0.3$)	at 30k, 45k and 55k	at 50k, 58k and 62k
Total steps / batch size	60,000 / 16	66,000 / 16
Precision	Mixed precision (AMP)	
Validation / checkpoint interval	Every 1,000 / 2,000 steps	
Hardware and time	NVIDIA T4 (Kaggle), about 3.6 h	NVIDIA T4 (Colab up to 44k, then Kaggle), about 4 h
Seed	1234 (data selection and training)	

Table 4.1: Training configuration

4.2 Training Loss

Figures 4.1 and 4.2 show the training losses: the total loss, the mel loss before and after the PostNet, and the pitch, energy and duration losses.

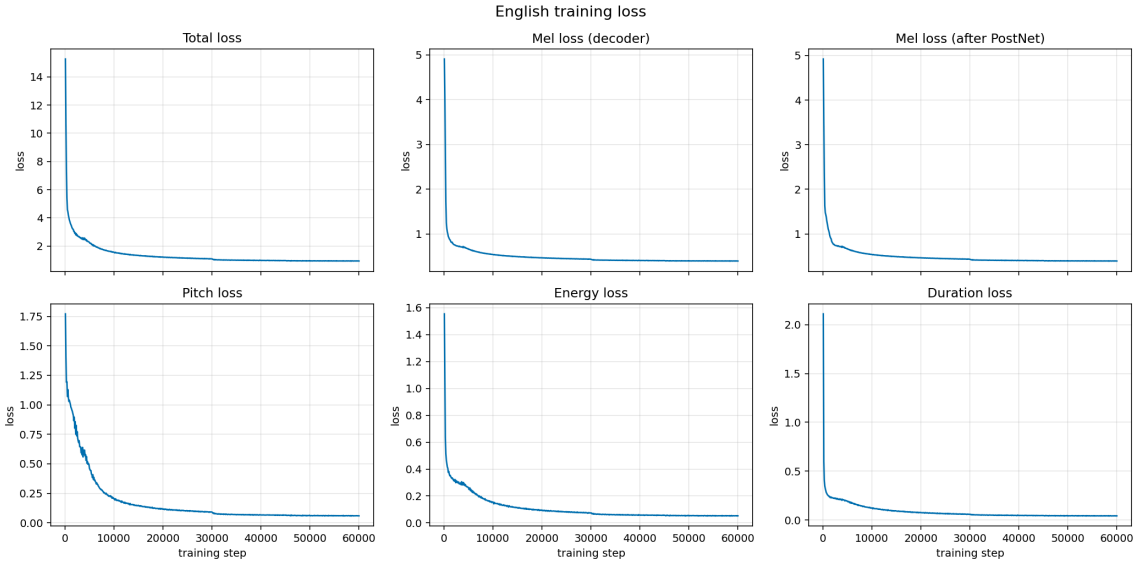


Figure 4.1: English training loss

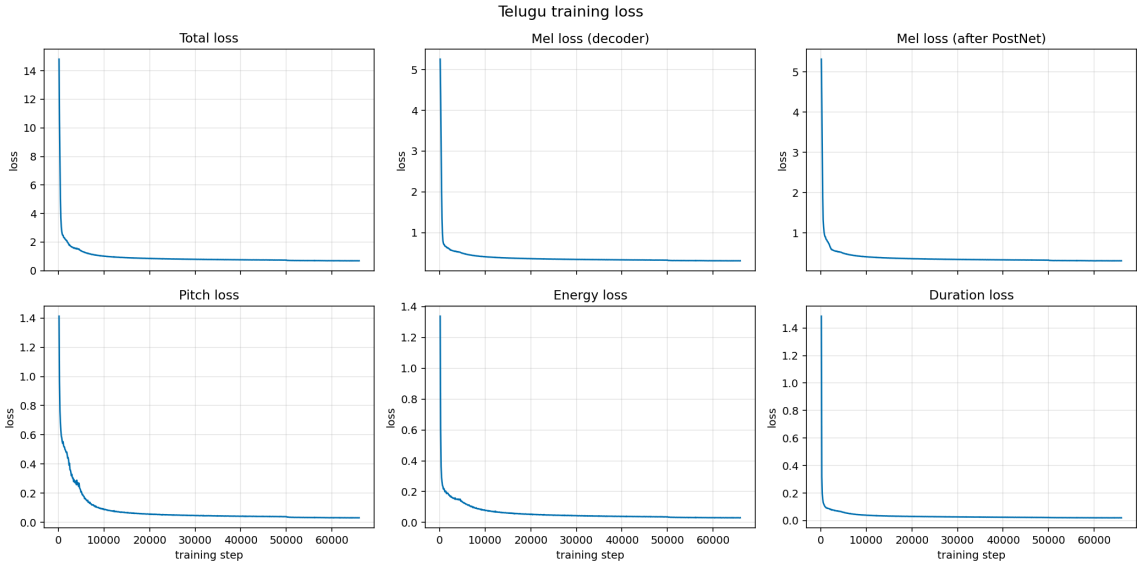


Figure 4.2: Telugu training loss

Both runs converge smoothly. The total loss falls from about 15 at the start to 0.95 for English and 0.69 for Telugu. Around step 4k, where the learning-rate warm-up ends, the decrease briefly slows down, and there are small drops after the first decay step (30k for English and 50k for Telugu). There are no loss spikes, and the Telugu curve shows no break at step 44k, where training moved from Colab to Kaggle.

4.3 Validation Loss, Convergence and Overfitting

Figures 4.3 and 4.4 show the validation losses. The orange curve is the mel loss when the model uses its own predicted pitch and energy instead of the measured values, and the red line marks the step at which `best.pt` was saved. Table 4.2 lists, for each loss, the lowest validation value with its step, the final validation value and the final training value, averaged over the last 2k steps.

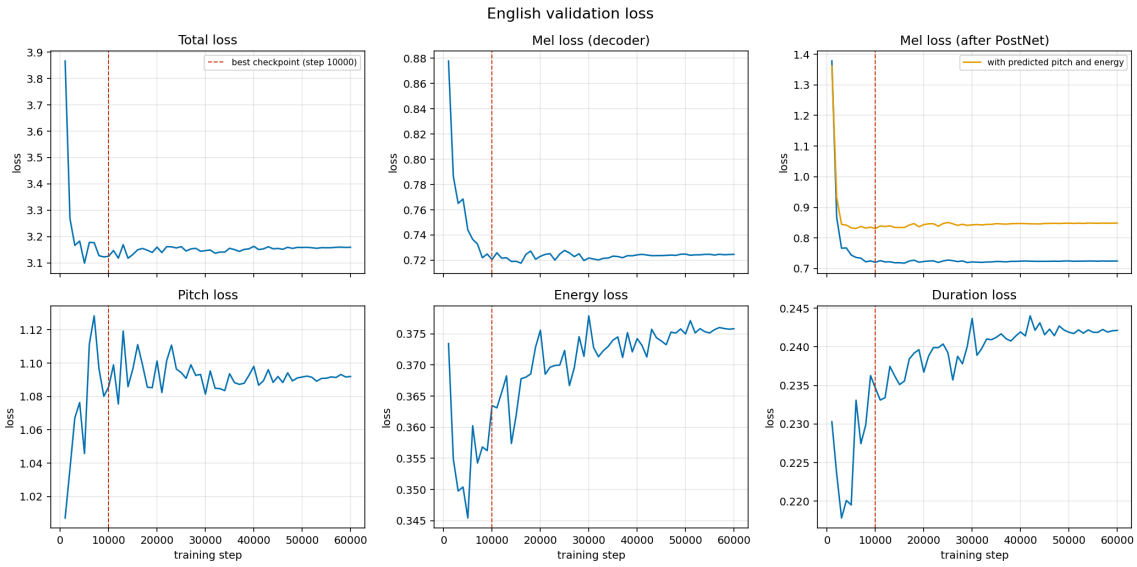


Figure 4.3: English validation loss

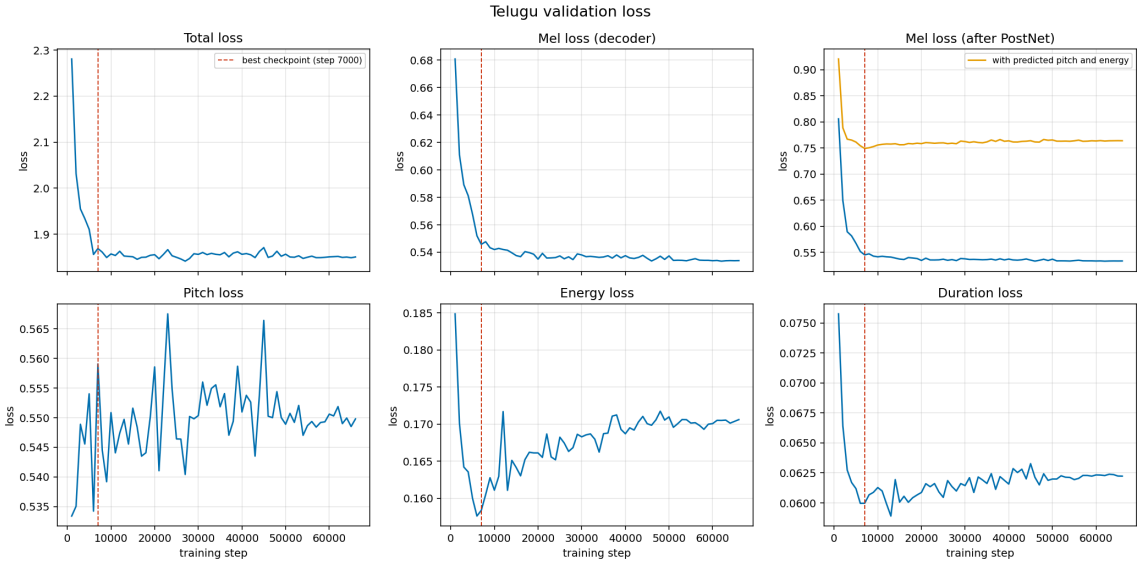


Figure 4.4: Telugu validation loss

Loss	English			Telugu		
	min. (step)	**final**	**train**	**min. (step)**	**final**	**train**
Mel after PostNet	0.717 (16k)	0.725	0.395	0.533 (62k)	0.534	0.306
Mel, predicted pitch/energy	0.830 (10k)	0.848	–	0.749 (7k)	0.764	–
Pitch	1.007 (1k)	1.092	0.060	0.533 (1k)	0.550	0.031
Energy	0.345 (5k)	0.376	0.053	0.158 (6k)	0.171	0.031
Duration (log)	0.218 (3k)	0.242	0.043	0.059 (13k)	0.062	0.020

Table 4.2: Validation and training losses

The decoder converges quickly and does not overfit. Its validation mel loss reaches a plateau after about 10–16k steps and then stays flat. For English the final value is less than 1% above the minimum reached at 16k. For Telugu it stays at about 0.536–0.538 from 16k to 50k and improves slightly once the learning rate decays, reaching its minimum at 62k. The training mel loss keeps falling, so the gap between training and validation grows (0.40 against 0.72 for English, 0.31 against 0.53 for Telugu), but the validation loss does not get worse.

The variance predictors behave differently. Their validation losses reach a minimum early, between 1k and 13k steps, and then rise by 3–11% while the training losses go towards zero. This is mild overfitting. The final English pitch loss, for example, is 18 times higher on validation data than on training data.

English pitch is also hard to predict in the first place. The validation pitch targets have a variance of 1.23 for English and 1.04 for Telugu, so the final English model explains only 11% of the pitch variation, against 47% for Telugu. Energy is predicted much better, with 67% and 86% explained. A likely reason is that the accented Svarah speakers say the same words with very different intonation, so the text alone says little about English pitch.

Overall, the Telugu losses are clearly lower than the English ones: the validation mel loss by about a quarter, and the pitch, energy and duration losses by half or more. One consistent studio speaker is simply easier to model than 114 accented speakers with about 2.6 minutes of speech each.

4.4 Checkpoint Selection

During training, `best.pt` is saved at the lowest validation mel loss computed with predicted pitch and energy. This happened at step 10k for English and step 7k for Telugu. However, this loss rewards smooth, average output, so a low value does not guarantee the most intelligible speech. The two candidates, `best.pt` and `final.pt`, were therefore also compared on the generated speech itself, using the evaluation described in Chapter 5. Table 4.3 shows the result.

Measure	`best.pt`	`final.pt`
English test sentences, word errors over 7 voices (calibrated speed)	51	**41**
English test sentences, word errors of voice S006	8	**3**
English validation WER, all voices (uncalibrated)	13.9%	14.5%
Estimated quality (SQUIM PESQ), S006	3.78	**3.87**
Telugu test sentences, phone errors of `male_a`	1	1
Telugu validation PER, all session IDs	6.9%	6.9%
Estimated quality (SQUIM PESQ), `male_a`	4.01	**4.07**

Table 4.3: Comparison of `best.pt` and `final.pt` on generated speech

On the validation clips the two checkpoints are practically tied: the English difference of 0.6 points corresponds to about 5 words out of 892. On the English test sentences and on estimated quality, `final.pt` is clearly better. For Telugu the two are equally intelligible, and `final.pt` scores slightly higher on estimated quality. For this reason `final.pt` is the released checkpoint for both languages.

Chapter 5

Observations on the Generated Speech

5.1 Evaluation Method and Released Voices

Both models are multi-speaker models, so they can speak with any voice that was seen in training. The released voices were chosen by how well their test sentences were recognised. For English this is S006, a female speaker from Srinagar with an average pitch of about 208 Hz, whose test sentences had the fewest word errors. For Telugu it is `male_a`, which ties with `male_b` and is the session ID with the most data.

The English duration predictor turned out to predict durations that are too short: for the seven voices that were evaluated, the real durations are 8–27% longer than the predicted ones, and 11% longer over the whole validation set (Section 7.1). English synthesis therefore uses the duration control that FastSpeech 2 provides. The scale for each voice is the ratio of real to predicted duration on the validation set, which for S006 is $\times 1.078$ for speech and $\times 1.819$ for pauses. For Telugu the real speech durations are only about 2% longer than the predicted ones, so Telugu is not scaled; its pauses are under-predicted more strongly, with real pauses about 22% longer.

Intelligibility was measured with Whisper speech recognition [7], using `openai/whisper-small.en` for English and `vasista22/whisper-telugu-small` for Telugu. Telugu is scored by phone error rate (PER) instead of word error rate (WER). Telugu words are long, and the recogniser often splits or joins them differently from the text, so even the real recordings get 35% word errors. Table 5.1 shows the results on validation clips, together with two reference points: the real recordings, and the real mel-spectrograms passed through the vocoder.

Validation clips (unseen sentences)	**English WER (70 clips, 7 voices)**	**Telugu PER (48 clips, 4 IDs)**
Real recordings	9.9%	6.0%
Vocoder only (real mel → HiFi-GAN)	11.4%	6.4%
FastSpeech 2, `final.pt`	14.5%	6.9%
FastSpeech 2, `final.pt`, calibrated speed	12.6%	–

Table 5.1: Intelligibility on unseen validation sentences

The test sentences are new to the models as well. None of them occurs in the training or validation data, and many of their words never appear there either. In English these include *antidisestablishmentarianism*, *considerable*, *proceedings*, *orchestra*, *praised*, *evaluate*, *proposed*, *unexpectedly* and *andaroo*. In Telugu, 14 of the 41 words are unseen (counting numbers as spoken words), including the loanword in T1 and every word of T3 after *puli*. The results below therefore show how well the models generalise to new text.

Speech quality was estimated with TorchAudio-SQUIM [8], which predicts PESQ (on a 1–4.5 scale) and STOI (from 0 to 1) without needing a reference recording. Table 5.2 gives the values per sentence; columns 1–4 stand for E1–E4 and T1–T4. Every test sentence scores between 3.85 and 4.22 PESQ except E3, the sentence with the broken word, at 3.57, and STOI is at least 0.98 everywhere. These are model-based estimates and not the result of a listening test.

Sentence	**1**	**2a**	**2b**	**3**	**4**	**Mean**
English (S006), PESQ	3.97	3.93	4.00	3.57	3.85	3.87
English (S006), STOI	0.991	0.997	0.995	0.981	0.991	0.991
Telugu (`male_a`), PESQ	3.95	4.22	4.10	3.96	4.14	4.07
Telugu (`male_a`), STOI	0.997	0.998	0.999	0.999	0.998	0.998

Table 5.2: Estimated quality of each generated test sentence

5.2 English Test Sentences

Table 5.3 lists, for each English test sentence, what the front-end produced, what the recogniser heard and what stands out. The recogniser counted 3 errors in 58 words, and none of them is a real mispronunciation. Two come from the hyphen it writes in “anti-disestablishmentarianism”, and one from its spelling of the made-up name “andaroo”.

ID	Challenge	Front-end output	Recognised as	Observation
E1	Very long rare word	28 phones, from the CMU dictionary	"The anti-disestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."	All words correct; the 2 counted errors are the hyphen. The long word takes 1.94 s.
E2a	Heteronym (noun)	`K AA1 N D AH0 K T`	Same as the input	Correct; full first vowel, as in the noun.
E2b	Heteronym (verb)	`K AH0 N D AH1 K T`	Same as the input	Correct; reduced first vowel (35 ms), as in the verb.
E3	Broken word, kept as written	*rhyth* `R AY1 DH` + *m* `EH1 M`	"The rhythm of the orchestra changed unexpectedly."	The literal reading is still recognised as "rhythm"; the separate "m" gets the highest energy.
E4	Time, date, year and name	*ten forty five A M … twelve november twenty twenty six*; andaroo `AE0 N D AA1 R UW1`	"… at 10.45 AM on 12 November 2026, so Andarouche should arrive on time."	All numbers correct; only the unseen name is spelled differently.

Table 5.3: English test sentences (voice S006)

5.3 Telugu Test Sentences

For Telugu (Table 5.4) the recogniser got 1 of 284 phones wrong. In T1 it heard the short *a* of *-ma-* as a long *ā* and split the long loanword into two words there. Everything else, including all the numbers in T4, matched the text exactly.

ID	Challenge	Front-end output	Recognised as	Observation
T1	English loanword in Telugu script	`n y uu r oo ph aa r m a k oo l aa j i k a l`	న్యూరోఫార్మా కోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు	1 of 66 phones differs: a short *a* heard as long *ā*, where the loanword is split. The rest is exact.
T2a	పట్టు ‘grip’	`p a T T u`	Same as the input	Exact (0/31); the doubled consonant is held.
T2b	పట్టు ‘silk’	`p a T T u`	Same as the input	Exact (0/37); pronounced as in T2a, which is correct for this homograph.
T3	Broken word అడ విలో, kept as written	`a D a + v i l oo`	పులి అడవిలో నిశ్శబ్దంగా నడిచింది	Exact (0/31); no pause is inserted, so it is recognised as one word.
T4	Year, ordinal and clock time	రెండు వేల ఇరవై ఆరు, పన్నెండవ, పది నలభై ఐదు	The input, with every number written out in Telugu words	Exact (0/119); all numbers spoken correctly.

Table 5.4: Telugu test sentences (voice `male_a`)

5.4 Overall Speech Quality

Both voices are stable: no words are skipped or repeated, as one would expect from a model with explicit durations.

The English voice is modelled on S006, an Indian-English speaker whose first language is Kashmiri. Without calibration it speaks about 8% faster than she does, and calibration makes its total durations on the validation sentences match hers. On the test sentences it still speaks 13–16 phones per second, faster than her average of about 12.4 phones per second in the recordings. Its intonation is flatter than natural speech (Chapter 6). On listening, some parts of the English sentences are harder to follow than the recognition results suggest: besides the fast rate, some stressed vowels are very short, for example 35 ms for the stressed vowel of “forty” in E4 and 46 ms for the stressed EH1 in *antidisestablishmentarianism*. A recogniser can use the context to fill in such sounds, which a listener who hears the sentence for the first time cannot always do.

The Telugu voice keeps vowel length and doubled consonants (Chapter 6). Its intonation spans 12–16 semitones and falls clearly at the end of each sentence. The vocoder costs little. Real mel-spectrograms passed through the universal HiFi-GAN lose only 1.5 (English) and 0.4 (Telugu) points of intelligibility (Table 5.1), so the acoustic model, not the vocoder, limits the quality.

Chapter 6

Variance-Adaptor Analysis

6.1 Reading the Plots

The plots required by the assignment are stored in three subfolders, `duration`, `pitch` and `energy`, of `Results/<Language>/variance_adaptor_plots/`. Each plot shows the generated mel-spectrogram with word and phone boundaries and the predicted value for every phone. The pitch and energy plots also show the contour measured on the generated audio.
The measured contours follow the predictions closely. Per phone, the measured F0 differs from the predicted F0 by a median of 0.24 semitones in English and 0.26 semitones in Telugu, and 99% and 96% of the phones agree within 2 semitones. The largest difference, 4.7 semitones in T3, is a pitch jump at a word boundary. In other words, the decoder really produces the prosody that the variance adaptor predicts, so the predictions can be read as a description of the output. Matplotlib cannot draw Telugu conjuncts correctly, so the Telugu plots label words in ISO 15919 transliteration.
To save space, this chapter combines the three predictions for each sentence into one figure (Figures 6.1–6.10), drawn from the same values stored in `inference_details.json`. The top panel shows the phone durations as bars, with pauses in orange; the middle panel shows the predicted F0 of each phone; and the bottom panel shows the predicted energy of each phone in arbitrary units. Dotted lines mark word boundaries, Telugu words are transliterated in the same way, and English durations include the calibration factor. Table 6.1 summarises the prosody of all ten sentences. Pitch ranges are given both in hertz and in semitones (st); the lengths are those of the synthesised speech, without the 0.15 s of silence added at each end of the audio files; and the rates of E4 and T4 count only speech, without the comma pause.

ID	Length (s)	Phones	Phones/s	F0 first → last word (Hz)	F0 range (Hz)	Range (st)	Pause (ms)
E1	6.11	87	14.2	234 → 188	178–257	6.3	–
E2a	3.85	51	13.2	224 → 181	173–234	5.2	–
E2b	3.56	56	15.7	194 → 206	178–240	5.2	–
E3	2.66	37	13.9	223 → 196	189–235	3.8	–
E4	6.26	81	13.2	229 → 183	177–248	5.9	197
T1	4.30	66	15.4	222 → 142	113–286	16.1	–
T2a	2.08	31	14.9	221 → 146	119–243	12.4	–
T2b	2.19	37	16.9	184 → 142	117–251	13.2	–
T3	1.96	31	15.8	258 → 161	126–298	15.0	–
T4	7.52	120	16.4	217 → 124	112–258	14.5	279

Table 6.1: Prosody summary of the generated test sentences (predicted values)

6.2 English Test Sentences

6.2.1 E1: Long Rare Word

"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

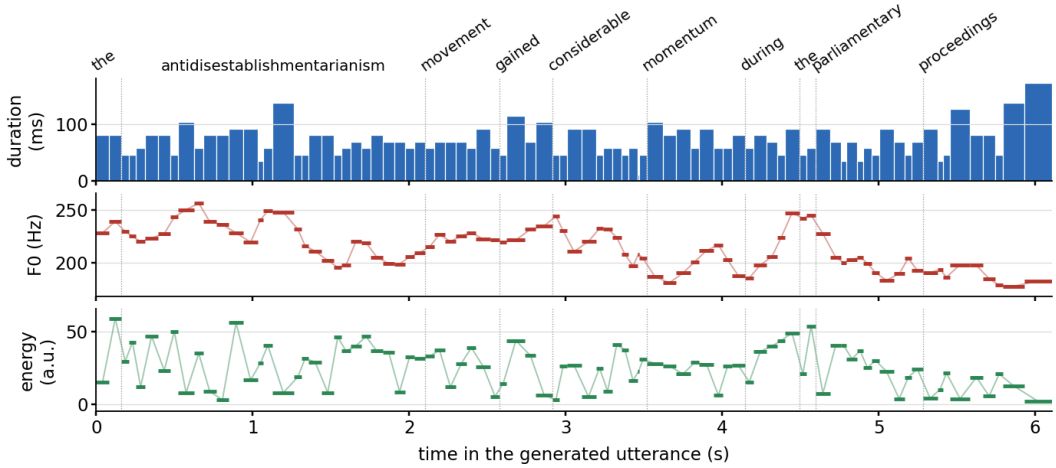


Figure 6.1: E1, predicted duration, F0 and energy

The 28-phone word *antidisestablishmentarianism* takes 1.94 s, about 32% of the sentence. Its phones stay short (35–139 ms, with S and SH the longest), so the model keeps a steady rhythm instead of slowing down for the rare word. The longest phone of the sentence comes at the very end, the final Z of "proceedings" (174 ms), after an NG of 139 ms. This is ordinary phrase-final lengthening.

F0 starts at 234 Hz on "The" and peaks at 257 Hz in the early syllables of the long word. It then declines to 178–198 Hz on "proceedings". The syllable with primary stress, *-tar-* (EH1, 196 Hz), is not a pitch peak: the flat English pitch predictor puts the accent too early. In the long word the vowels reach energies of 30–56, while the voiceless stops and fricatives of the sentence stay close to zero (T 3–12, S 4–9). The last word has the lowest energy, as is usual at the end of a sentence.

6.2.2 E2a: "Conduct" as a Noun

"The orchestra's conduct during the ceremony was praised by the audience."

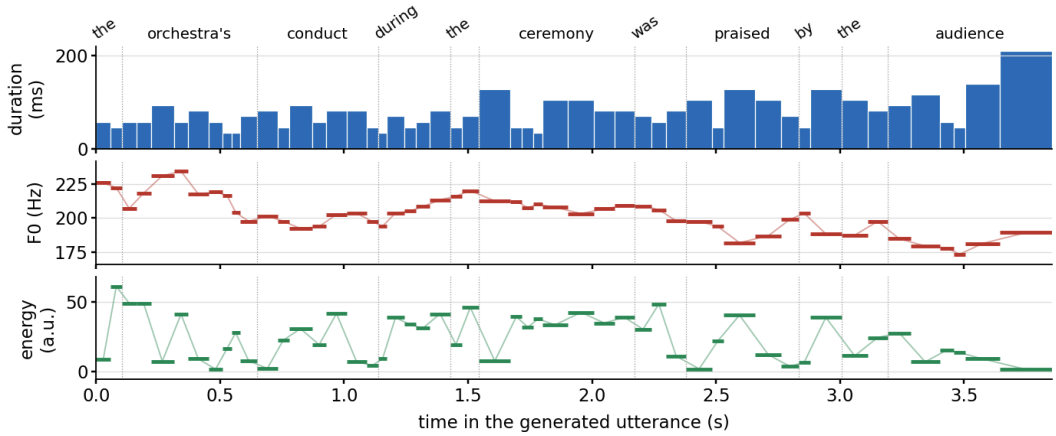


Figure 6.2: E2a, "conduct" as a noun

Here "conduct" is a noun, pronounced `K AA1 N D AH0 K T`. The stressed AA1 lasts 46 ms with an energy of 22, while the unstressed AH0 lasts 81 ms with an energy of 42, and F0 is almost level at 192–203 Hz. The noun reading therefore comes from the full first vowel and not from any prominence on the first syllable. The longer and louder second syllable is probably lengthening before the phrase boundary that follows "conduct".

Over the whole sentence F0 falls from 224 to 181 Hz. The last word, "audience", is lengthened, with its final S as the longest phone of the sentence (209 ms), and it has the lowest energy.

6.2.3 E2b: “Conduct” as a Verb

“The researchers will conduct an experiment to evaluate the proposed method.”

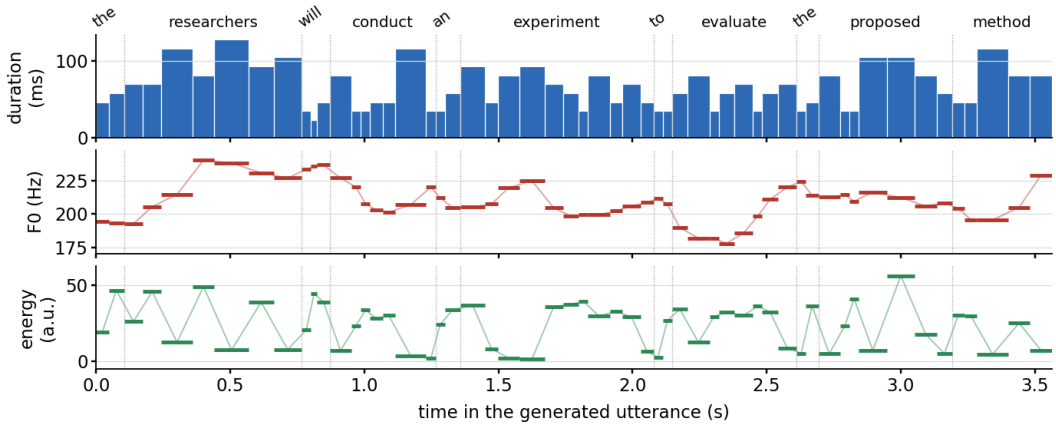


Figure 6.3: E2b, “conduct” as a verb

As a verb, “conduct” is pronounced `K AH0 N D AH1 K T`. The first vowel is reduced to 35 ms (energy 24), and the stressed AH1 is longer and louder (46 ms, energy 30), which is the expected pattern for the verb. The whole word lasts 395 ms here, against 488 ms as a noun in E2a. The highest word-level F0 (235 Hz) falls on “will”, just before the verb.

E2b is the fastest English sentence, at 15.7 phones per second. It is also the only sentence without an overall fall in pitch (194 → 206 Hz), because it ends with a slight rise to 229 Hz on the final D of “method”.

6.2.4 E3: Broken Word

“The rhyth m of the orchestra changed unexpectedly.”

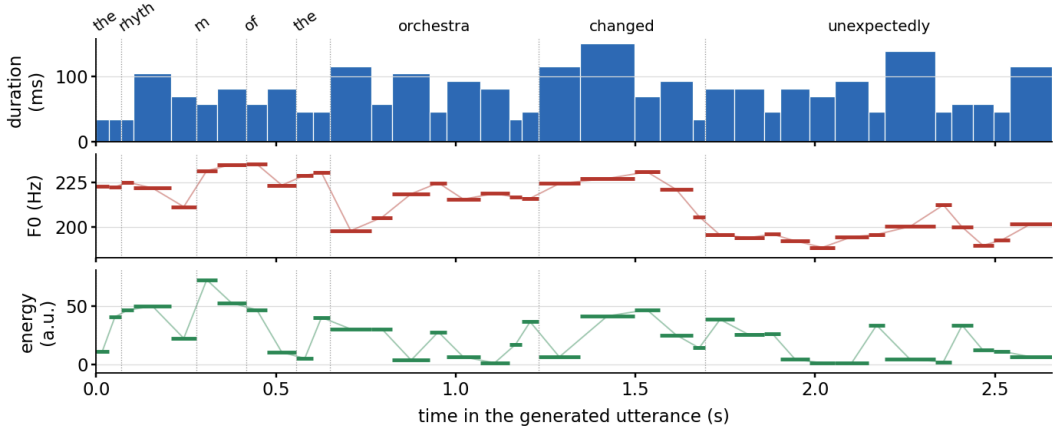


Figure 6.4: E3, broken word “rhyth m” kept as written

The space splits “rhythm” into “rhyth” (`R AY1 DH`) and “m” (`EH1 M`), which are read as two separate words. The isolated letter gets the highest energy in the sentence (72.5) and the highest word-level F0 (233 Hz), so the variance adaptor treats it like a stressed word. There is no punctuation, so no pause is inserted, and the two parts run together and are recognised as “rhythm”.

The longest phone in the sentence is the diphthong of “changed” (151 ms). The final word, “unexpectedly” (964 ms), ends with a lengthened IY0 (116 ms).

6.2.5 E4: Time, Date, Year and Name

> *“The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.”*

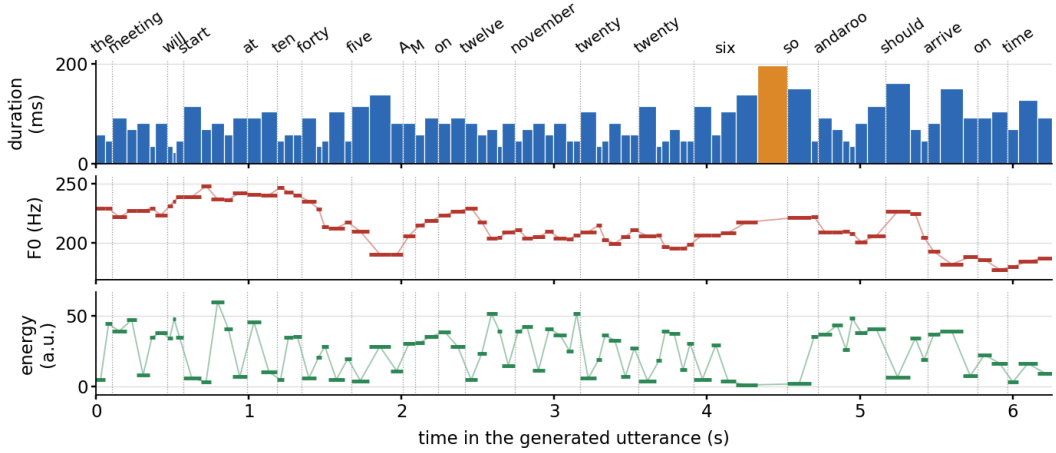


Figure 6.5: E4, time, date, year and name

Over “ten forty five”, F0 falls from 244 to 190 Hz, and “five” is lengthened, with a diphthong of 139 ms. F0 rises again on the spelled-out letters (206–219 Hz), which are short: A lasts 81 ms and M 151 ms.

The comma produces the only pause in the sentence, 197 ms long (the orange bar in Figure 6.5). Just before it, “six” is lengthened to 418 ms and its energy drops to 10. After it, F0 resets to 222 Hz on “so”, which marks a clear phrase boundary.

The name “andaroo” (441 ms) is read with an almost level F0 of about 207 Hz, the highest word energy of the sentence (39, shared with “will”) and a long final UW1 (116 ms), much like an unfamiliar word without a clear accent. The sentence ends low on “time” (179–187 Hz, energy 10), and the overall fall is from 229 to 183 Hz.

6.3 Telugu Test Sentences

6.3.1 T1: English Loanword

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

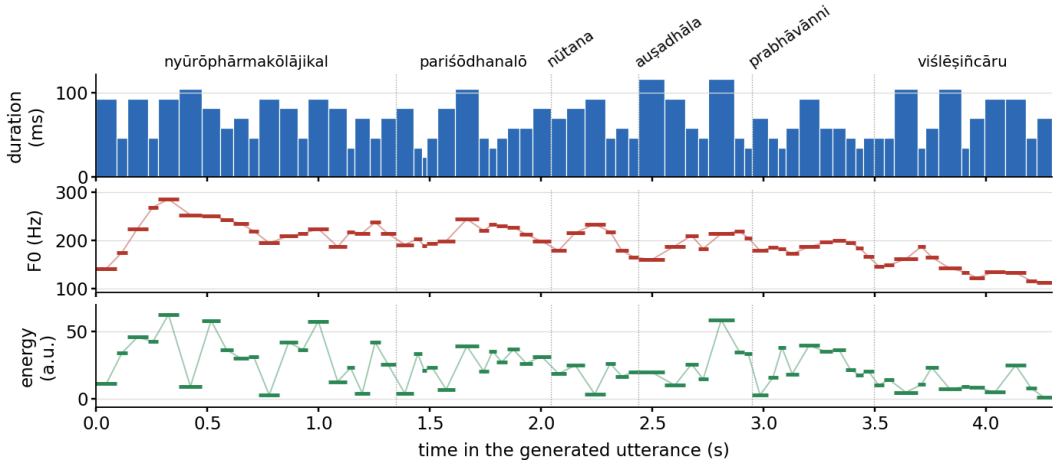


Figure 6.6: T1, long English loanword in Telugu

The loanword *nyūrōphārmakōlājikal* takes 1.35 s, about 31% of the sentence. Its F0 rises steeply from 142 Hz to the sentence maximum of 286 Hz on the second syllable (*rō*) and then falls back, which gives the first word a strong accent.

Long vowels (*ū*, *ō*, *ā*: 81–93 ms) are about twice as long as short *a* (46 ms), so the Telugu length contrast is kept. They also carry the energy peaks (*ō* 62, *ā* 57–58). The final verb *viślēṣiñcāru* falls to 113 Hz with energy near zero, and over the whole sentence F0 goes from 222 to 142 Hz.

6.3.2 T2a: *paṭṭu* Meaning ‘Grip’

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

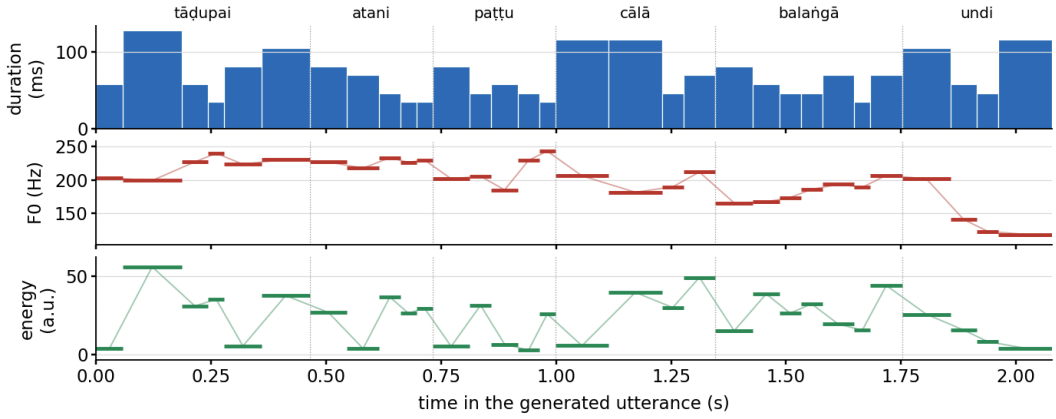


Figure 6.7: T2a, *paṭṭu* meaning ‘grip’

The doubled consonant *ṭṭ* becomes two T phones (58 and 46 ms), 104 ms in total, clearly longer than the single *t* of *tāḍupai* (58 ms) and *atani* (70 ms), so the doubling is kept. F0 dips to 185 Hz during the closure and rises to 243 Hz on the final *u*. The highest word-level F0 is on *atani* (227 Hz), and the final *undi* falls to 119 Hz with a lengthened final *i* (116 ms).

6.3.3 T2b: *paṭṭu* Meaning ‘Silk’

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

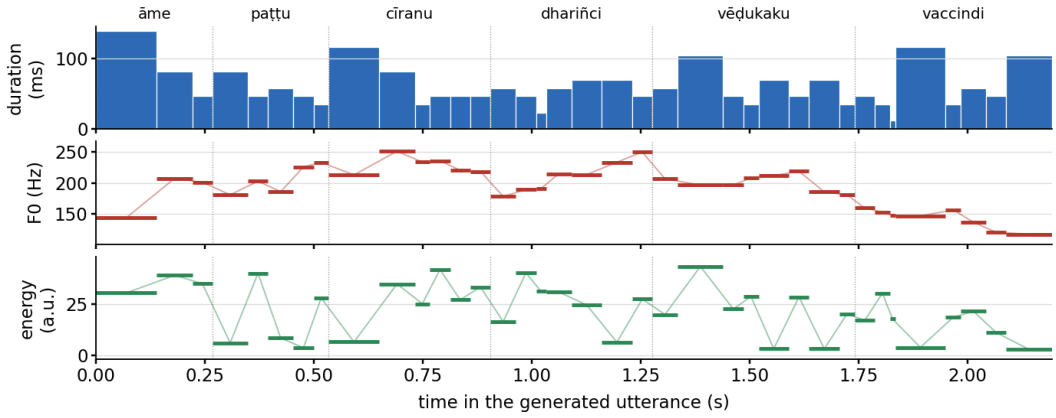


Figure 6.8: T2b, *paṭṭu* meaning ‘silk’

In this sentence *paṭṭu* means silk, but it is spelled and pronounced exactly as in T2a. Its phone durations are identical to those in T2a (81, 46, 58, 46 and 35 ms), and its pitch

contour is nearly the same (181–233 Hz against 185–243 Hz). Giving the homograph the same pronunciation in both meanings is correct for Telugu. The slightly lower pitch fits its earlier position in the sentence: it is the second word here and the third word in T2a.
The doubled *cc* of *vaccindi* gives a closure of 128 ms. F0 peaks on *cīranu* (229 Hz) and falls to 117 Hz at the end. At 16.9 phones per second, T2b is the fastest of all ten sentences.

6.3.4 T3: Broken Word

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

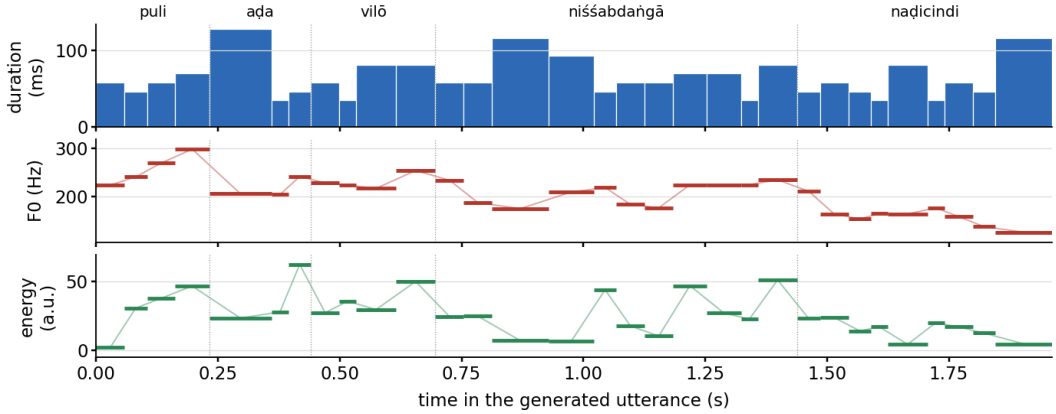


Figure 6.9: T3, broken word అడ విలో kept as written

Here the space splits *aḍavilō* into two prosodic words. *aḍa* starts with a lengthened *a* (128 ms) and ends with an F0 jump to 241 Hz and the highest energy in the sentence (62). *vilō* then gets its own final rise (254 Hz). No pause is inserted, so the speech still flows as one word and the recogniser hears అడవిలో.
The doubled *śś* lasts 209 ms. T3 starts at the highest F0 of all the test sentences (*puli*, 258 Hz) and declines to 126 Hz.

6.3.5 T4: Year, Ordinal and Clock Time

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

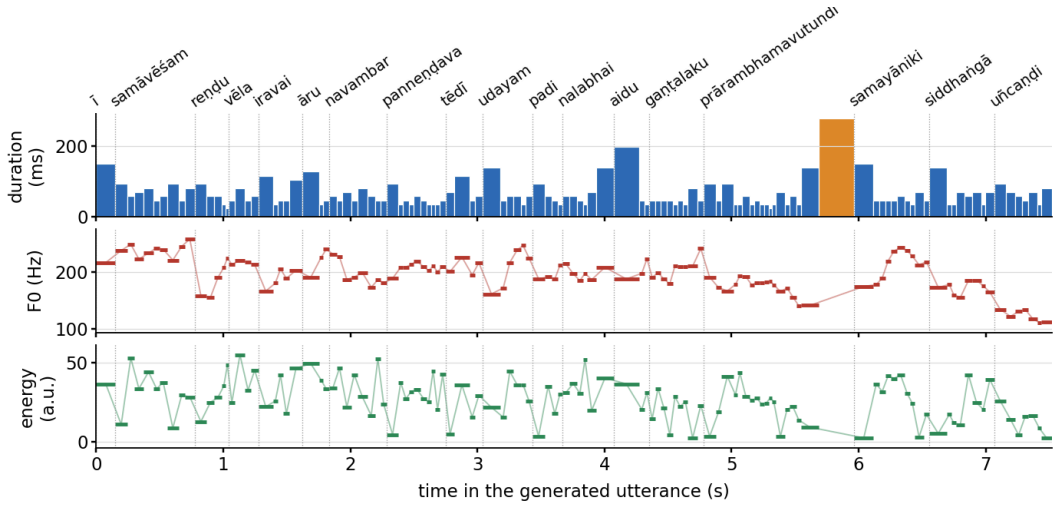


Figure 6.10: T4, year, ordinal and clock time

All numbers are spoken as words: 2026 becomes *reṇḍu vēla iravai āru*, 12వ becomes *panneṇḍava* and 10:45 becomes *padi nalabhai aidu*. Each number word has its own rising contour (*reṇḍu* 156→225 Hz, *āru* 191→241 Hz, *aidu* 188→224 Hz), a pattern typical of careful list reading. The diphthong *ai* in *aidu* is the longest phone of the sentence (197 ms). The comma gives the only pause, 279 ms long (the orange bar in Figure 6.10). F0 drops to about 140 Hz at the end of *prārambhamavutundi* and resets to about 240 Hz on *samayāniki*. The sentence ends with *uñcaṇḍi* at 112 Hz and almost no energy.

6.4 Patterns Across Sentences

In 9 of the 10 sentences, all except E2b, F0 falls from the first word to the last, by 26–46 Hz in English and 42–97 Hz in Telugu, and the last word has the lowest energy. The last phones of a sentence or phrase are among the longest (Z 174 ms in E1, S 209 ms in E2a, and the final *i* of 104–116 ms in T2a, T2b and T3).

Pauses appear only at punctuation, at the commas of E4 and T4, and each is followed by a pitch reset; the broken words in E3 and T3 change the word-level prosody but add no pause. Energy follows sonority: it is high on vowels and nasals and close to zero on voiceless stops and fricatives. In Telugu, long and short vowels, and single and doubled consonants, are kept clearly apart.

The two languages differ most in pitch range: 12–16 semitones in Telugu against only 4–6 in English, whose intonation is much flatter because its pitch is hard to predict (Section 4.3).

Chapter 7

Comparison, Limitations and Improvements

7.1 English versus Telugu

Table 7.1 puts the main numbers side by side. The share of the pitch and energy variance that the predictors explain, the duration bias and the gap to real speech are all measured on the validation set; the test-sentence errors are those of the released voices.

Aspect	**English**	**Telugu**
Training data	4.99 h, 114 speakers (≈2.6 min each)	5.00 h, one studio speaker
Phone set and alignment	69 ARPAbet phones, pretrained aligner	43 phones, aligner trained on the data
Validation mel loss	0.725	0.534
Pitch / energy explained	11% / 67%	47% / 86%
Duration bias	real 8–27% longer for the evaluated voices (calibrated)	real about 2% longer (not scaled)
Gap to real speech	+2.7 points WER	+0.9 points PER
Test-sentence errors	3/58 words, none real	1/284 phones
Test-sentence pitch range	4–6 semitones	12–16 semitones

Table 7.1: English and Telugu models compared

The Telugu model is clearly the stronger of the two, and the reasons are not hard to find. It learns one consistent studio speaker, its script maps almost directly to pronunciation, and its aligner was trained on exactly this voice. The English model has to learn 114 accented speakers from very little data per speaker. Its predictors see conflicting targets for the same text and fall back towards the average, which results in flat pitch and durations that are too short. Even so, its gap to real speech is fairly small, and the words of its test sentences come out right.

7.2 Limitations and Failures on the Test Sentences

- **E1:** the long word is pronounced correctly, but its pitch accent comes early instead of on the stressed syllable *-tar-*. This follows from the weak English pitch predictor.
- **E2a and E2b:** both pronunciations are correct, but only the verb shows its stress in duration and energy. The noun's stress shows only in its full first vowel.
- **E3 and T3:** the broken words are read literally, as the assignment requires. The separate "m" is stressed and *aḍa vilō* gets two word-final rises. A listener may notice this, even though the recogniser restores the intended words.
- **E4:** the unseen name "andaroo" depends on a predicted pronunciation, `AE0 N D AA1 R UW1`, and has no clear accent.
- **T1:** the loanword is read from its Telugu spelling without any knowledge of English stress. Its first syllables get an exaggerated pitch peak (286 Hz), and the recogniser splits the word in two.
- **English clarity:** the English speech is fast and some stressed vowels last only 35–46 ms, so parts of the English sentences are hard to follow by ear, even where the recogniser gets every word.
- **Coverage:** the English voice is modelled on a single accented speaker, the Telugu model covers only one male speaker, and the vocoder was not fine-tuned.

7.3 Possible Improvements

- **More consistent English data.** More data per English speaker, or fewer and cleaner speakers, together with utterance-level prosody conditioning, would give the pitch predictor something consistent to learn.
- **A better duration loss.** The English durations could be corrected inside the model with a loss on linear durations or a learned bias term. An MSE loss on log-durations predicts something close to a geometric mean, which lies below the arithmetic mean when durations vary a lot. This is a plausible cause of the English under-prediction.
- **Separate regularisation of the predictors.** The variance predictors overfit after 1–13k steps, so they could be stopped earlier or regularised more strongly than the decoder.
- **Smaller fixes.** Switching on the broken-word repair (already implemented for English), adding a loanword lexicon for Telugu, and fine-tuning HiFi-GAN on each dataset.

Chapter 8

Reproducibility

8.1 Software and Hardware

The code runs on Python 3.13 with PyTorch 2.10.0, torchaudio 2.11.0, NumPy 2.3.5, SciPy 1.17.1, librosa 0.11.0, soundfile 0.13.1, pandas 3.0.3, pyarrow 25.0.0, praat-parselmouth 0.4.7, tgt 1.5, g2p-en 2.1.0, NLTK 3.10.0, PyYAML 6.0.3 and Matplotlib 3.10.8. All of them are listed in `requirements.txt`. The Montreal Forced Aligner 3.4.1 is only needed for preprocessing and runs in its own conda environment.
Training ran on a single NVIDIA T4 GPU on Google Colab and Kaggle. Inference takes less than a minute per language on a CPU.

8.2 Configuration Files

`English/config.yaml` and `Telugu/config.yaml` contain every setting used in this report. Apart from the language name, each file has six sections: `paths` (dataset, work and checkpoint folders, vocoder), `data` (selection filters, quality weights, split sizes, seed), `preprocessing` (audio, STFT, mel, pitch and energy), `model` (Table 2.1), `train` (Table 4.1) and `inference` (voice, English duration scales and the five test sentences exactly as given). Relative paths are resolved from the location of the configuration file.

8.3 Commands

The commands below rebuild everything from the raw datasets, which are expected in `Svarah/data` and `IndicTTS_Telugu/data`. Training should be given a new `--ckpt_dir`, as shown. Without it, training would append to the released loss logs and overwrite `final.pt`. An interrupted run resumes from `last.pt` in the same folder.

```
pip install -r requirements.txt
conda create -n aligner -c conda-forge montreal-forced-aligner=3.4.1
mfa model download acoustic english_us_arpa
mfa model download dictionary english_us_arpa
MFA=~/Documents/MFA/pretrained_models   ENV=<path-to-conda-envs>/aligner

python English/preprocess.py analyze
python English/preprocess.py select --mfa_dictionary $MFA/dictionary/english_us_arpa.dict
python English/preprocess.py align --mfa_env $ENV --mfa_acoustic_model
    $MFA/acoustic/english_us_arpa.zip
python English/preprocess.py features
python Telugu/preprocess.py select
python Telugu/preprocess.py align --mfa_env $ENV
python Telugu/preprocess.py features

python English/train.py --ckpt_dir English/checkpoints_retrained
python Telugu/train.py --ckpt_dir Telugu/checkpoints_retrained

python -B English/inference.py
python -B Telugu/inference.py
```

The two inference commands write the test audio, the variance-adaptor plots and the loss plots to `Results/<Language>/`.

8.4 Repository Layout

- `English/` and `Telugu/`: `train.py`, `preprocess.py`, `inference.py`, `config.yaml` and `checkpoints/`, which holds `final.pt` and the loss logs.
- `fastspeech2/`: the shared model, text front-ends, preprocessing, training, inference and plotting code, together with the HiFi-GAN generator weights.
- `Results/`: for each language, the training and validation loss plots, the five test audios, the duration, pitch and energy plots, and `inference_details.json`.
- `Report/`: this report.
- **Checkpoints:** the two `final.pt` files are larger than GitHub's file-size limit, so they are hosted on Google Drive, in the subfolders `English` and `Telugu`. Both checkpoints are available at:
  https://drive.google.com/drive/folders/1KYR0W8YtH0LEsdu4d1INNzAjR7Ino1uo

Chapter 9

Conclusion

In this work FastSpeech 2 was adapted from the reference implementation and trained separately for Indian-accented English and for Telugu. Each model used 5 hours of training audio and 1 hour of validation audio, with no sentence shared between the two sets.

Telugu support needed a rule-based front-end with number normalisation, an aligner trained on the Telugu data and separate speaker IDs for the recording sessions. The noisy, multi-speaker English corpus needed quality-based data selection, along with text normalisation and the handling of heteronyms.

Both models trained stably. The decoders converged after about 10–16k steps, and the variance predictors overfitted mildly after that. The released `final.pt` checkpoints were chosen by evaluating the generated speech, and not only by a loss value.

Both voices speak all ten test sentences with almost no recognition errors, including the heteronyms, numbers, dates, times, the long rare word, the unseen name and the broken words. The Telugu model comes within 0.9 points of real recordings. The English model is limited mainly by the small amount of data per speaker, and its fast, flat delivery makes parts of it hard to follow by ear. The variance-adaptor analysis shows that both models have learned to lower the pitch over a sentence, to lengthen phrase-final phones and to pause at punctuation, and the Telugu model also keeps vowel length and doubled consonants.

The clearest weakness is the flat English intonation, which comes from the low predictability of English pitch in this data. More consistent English data, or conditioning on utterance-level prosody, would be the most useful next step.

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